

SOFTWARE REQUIREMENTS & SYSTEM ARCHITECTURE SPECIFICATION

Household Service Management System (HSMS)

A Comprehensive Academic Case Study

<u>DEVELOPMENT LIFECYCLE</u>	<u>TARGET ARCHITECTURE</u>	<u>PRIMARY STACK</u>
Waterfall SDLC Model	Web-Based Centralized Platform	Java Spring Boot, MySQL, Bootstrap

1. EXECUTIVE INTRODUCTION

The Household Service Management System (HSMS) is a comprehensive web-based application designed to automate, streamline, and modernize the delivery of domestic maintenance and repair services. Traditionally, coordinating household services such as plumbing, electrical work, carpentry, and home cleaning relies heavily on manual interventions. Customers typically depend on physical directories, informal recommendations, or unorganized phone inquiries to find technicians. These conventional approaches are highly inefficient, time-consuming, and frequently result in scheduling conflicts, pricing discrepancies, and safety concerns due to unverified personnel.

The proposed Household Service Management System establishes a centralized digital platform that seamlessly connects administrators, professional service providers, and customers (homeowners). The system significantly minimizes manual effort, ensures accurate service tracking, guarantees transparent pricing, and provides real-time updates regarding service requests. It adheres strictly to the Waterfall Software Development Life Cycle (SDLC) model, guaranteeing a systematic and structured development process through sequential phases: requirement analysis, system architecture design, implementation, rigorous testing, deployment, and ongoing maintenance.

2. PROBLEM STATEMENT

Many urban residential communities, local service agencies, and individual homeowners continue to rely on fragmented and manual methods for managing essential household services. Service requests, provider assignments, appointment scheduling, billing, and customer feedback are largely maintained via physical ledgers, scattered spreadsheets, or informal messaging applications. These outdated processes are not only cumbersome to maintain but are also highly susceptible to human errors, including duplicate bookings, lost appointment records, inaccurate cost estimations, and prolonged response times.

As the demand for on-demand home maintenance accelerates, service agency administrators face mounting challenges in effectively tracking the real-time availability of their technicians, verifying worker credentials, managing prompt dispute resolution, and generating coherent operational reports. Simultaneously, customers experience profound frustration due to unnecessary delays in securing reliable professionals, opaque pricing structures, and a lack of accountability for substandard work. To resolve these operational bottlenecks, a robust, computerized Household Service Management System is indispensable. It aims to completely automate service workflows, elevate operational efficiency,

enforce security through rigorous provider verification, eradicate paperwork, and grant real-time transparency to all system stakeholders.

3. SYSTEM OBJECTIVES

- **Automate Service Workflows:** Digitally streamline the end-to-end process of booking, provider assignment, and service fulfillment tasks.
- **Minimize Manual Interventions:** Transition entirely from scattered contact logs and manual scheduling to a highly secure, centralized database platform.
- **Accurate Record Keeping:** Maintain precise, continuously updated customer profiles, verified technician credentials, service histories, and financial transaction records.
- **Streamline Scheduling & Billing:** Facilitate intuitive appointment scheduling with automated, transparent billing computations devoid of hidden charges.
- **Online Tracking & Quality Control:** Provide a structured mechanism for service tracking, transparent ratings, and customer reviews to ensure high service standards.
- **Comprehensive Reporting:** Empower platform administrators with action-oriented operational reports detailing revenue, service completion rates, and provider performance.

4. SCOPE OF THE PROJECT

The Household Service Management System is explicitly tailored for residential complexes, localized home service agencies, and independent homeowners seeking reliable maintenance solutions. The system's functional boundaries encompass robust Customer Registration and Authentication, comprehensive Service Provider (Technician) Onboarding, and meticulous Admin Dashboards.

The platform supports a wide array of designated Service Categories, specifically including: Electricians, Plumbers, Carpenters, Cleaners, Painters, AC Repair Specialists, Appliance Repair Technicians, Pest Control Experts, and Gardening Professionals. Furthermore, the scope encompasses end-to-end Booking Management, flexible Appointment Scheduling, secure Online Payment gateways, real-time Service Tracking, automated Notifications (Email/SMS), dynamic Ratings & Reviews, exhaustive Administrative Report Generation, and structured User Profile Management.

5. COMPARATIVE ANALYSIS: EXISTING VS. PROPOSED SYSTEM

Limitations of the Existing System: The prevailing manual paradigm is crippled by excessive dependencies on phone-based scheduling, resulting in high data redundancy, overlapping appointments, and human errors in billing calculations. It is fundamentally vulnerable to security and safety risks, as deploying unverified service personnel poses a direct threat to homeowners. Furthermore, it suffers from delayed grievance resolution and a profound lack of a centralized data repository to track historical service trends.

The Proposed System: The web-based HSMS guarantees highly secure, role-based access meticulously segregated for Administrators, Service Providers, and Customers. Featuring intuitive,

dynamic dashboards powered by Java Spring Boot, real-time MySQL database connectivity, automated appointment tracking, and instant access to performance metrics, it establishes an exceptionally agile, transparent, and responsive domestic service environment.

6. FUNCTIONAL REQUIREMENTS

- **Authentication Module:** Secure registration and login for Admins, Service Providers, and Customers utilizing encrypted passwords, Spring Security configurations, JSON Web Tokens (JWT) for session management, and robust logout handling.
- **User Profile Management:** Comprehensive CRUD operations for user profiles, empowering customers and providers to seamlessly update their personal information, contact details, and service addresses.
- **Service Category Management:** An exclusive administrative module dedicated to dynamically configuring, categorizing, and pricing various services (e.g., Electrician, Plumber, Pest Control).
- **Booking & Appointment Scheduling:** An interactive engine allowing customers to seamlessly browse available services, select preferred dates and time slots, and confirm appointments based on regional provider availability.
- **Online Payment System:** Integration of secure digital payment processing to facilitate upfront or post-service transactions, complete with automated invoice generation and outstanding balance tracking.
- **Service Tracking Module:** A real-time tracking interface where customers can monitor the exact status of their service requests (e.g., Pending, Provider Assigned, In Transit, Completed).
- **Notification Engine:** Automated alerts dispatched to users regarding booking confirmations, provider assignment updates, payment receipts, and upcoming appointment reminders.
- **Ratings & Reviews System:** A transparent feedback mechanism enabling customers to rate completed services and write detailed reviews, directly impacting a provider's overall platform visibility and standing.
- **Admin Dashboard & Report Generation:** A sophisticated control panel for administrators to export graphical and tabular analytics covering daily revenue, service volume, category popularity, and pending disputes.

7. NON-FUNCTIONAL REQUIREMENTS

Quality Attribute	Requirement Specification
Security	Implements robust role-based authorization via Spring Security, stringent password hashing, secure JWT stateless sessions, and definitive input validation to thwart SQL injection and Cross-Site Scripting (XSS).
Performance	Engineered for rapid server response times (<2 seconds) and optimized database queries capable of sustaining high concurrent multi-user interactions during peak service booking hours.
Reliability	Ensures exceptionally high system stability fortified by automated daily MySQL database backups, uncompromising data integrity constraints, and reliable continuous uptime.
Scalability	Constructed on a modular architecture expressly designed to seamlessly incorporate new service categories, expansive geographic zones, and exponentially higher user bases without performance degradation.
Usability	Features an intuitively clean, highly responsive user interface meticulously crafted with Bootstrap 5, demanding minimal technical literacy from homeowners and field technicians.
Maintainability & Portability	Leverages a rigidly structured, modular codebase adhering to MVC architecture, facilitating effortless maintenance, cross-browser compatibility, and unparalleled deployment flexibility across diverse cloud environments.

8. FEASIBILITY ANALYSIS & TARGET USERS

Technical Feasibility: Highly achievable utilizing robust, industry-proven enterprise technologies (Java Spring Boot, MySQL, HTML5/CSS3/JavaScript).

Economic Feasibility: Exceptionally cost-effective owing to the utilization of open-source frameworks and zero licensing fees associated with the primary development stack.

Operational Feasibility: Significant user acceptance is anticipated, driven by a simplified, modern user interface designed specifically for rapid service discovery and booking.

Schedule Feasibility: A structured, sequential planning approach under the Waterfall SDLC guarantees meticulous phase completion aligned strictly with academic submission timelines.

Primary System Stakeholders: (1) Administrators: Oversee full system configuration, provider verification, dispute resolution, and financial oversight. (2) Service Providers: Manage daily schedules, update job statuses, and receive digital payments. (3) Customers: Browse service catalogs, schedule appointments, track statuses, and submit ratings.

9. PROPOSED TECHNOLOGY STACK

Frontend	HTML5, CSS3, Bootstrap 5, JavaScript
Backend	Java Spring Boot (MVC Architecture)
Database	MySQL Relational Database Management System
Security & Tools	Spring Security, JSON Web Tokens (JWT), Maven, Git

10. EXPECTED OUTCOMES & FUTURE SCOPE

The comprehensive implementation of the Household Service Management System will permanently replace archaic, paper-based, and uncoordinated service scheduling workflows. It will drastically reduce human error, optimize technician allocation, guarantee secure financial transactions, drastically improve provider accountability, and supply platform administrators with precise, data-driven operational insights.

Conclusion & Future Enhancements: The HSMS delivers an undeniably robust, scalable, and highly secure foundation for domestic service administration developed via the Waterfall SDLC. Future phases can expand aggressively upon this baseline by integrating real-time GPS tracking for field technicians, AI-driven dynamic pricing algorithms based on demand, Augmented Reality (AR) tools for remote issue diagnosis, and dedicated native mobile applications (iOS/Android) for enhanced on-the-go accessibility.